

# **Project Proposal**

**ICT 222-2**

## **Phishing Detection System**

Group No: 05

| Registration Number | Name                        |
|---------------------|-----------------------------|
| UWU/ICT/23/004      | K. A. N. P. Perera          |
| UWU/ICT/23/005      | L. V. C. S. U. Kumari       |
| UWU/ICT/23/026      | S. H. Thebuwana             |
| UWU/ICT/23/080      | A. W. D. H. T. Jayawickrama |
| UWU/ICT/23/081      | S. A. N. Rathnasinghe       |

Supervised By:  
Mr. V. Dishankan

**Department of Information and Communication Technology**

**Faculty of Technological Studies**

**Uva Wellassa University**

**2023/2024 Batch**

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## 1. Introduction

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The exponential development of internet technology and related services has led to an increase in cyber security threats worldwide. One of the most serious kinds of cyber-attacks includes phishing attacks that involve creation of false websites and URLs in order to steal confidential information from the victim including password and bank credentials. Because of the high level of sophistication of these attacks, users are often unable to tell which websites are safe and which are malicious. Typical examples of phishing attacks include the use of false banking websites, login pages, emails, and social engineering methods that can cause financial losses to the victims along with identity theft. With the development of online financial transactions and digital communication, there is a greater need for a better phishing detection system. In order to address this issue, a Multilayer Phishing Detection System based on the Java platform will be developed for analysis of URLs and detection of suspicious websites. The proposed system will make use of rule-based phishing detection methods along with external threat intelligence APIs for URL analysis and phishing risk assessment.

## 2. Background and Motivation

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Several approaches have been proposed to detect phishing scams and prevent such threats from occurring in the first place. There are already a few security solutions that can address the issue through browser warnings, blacklisting approaches, and paid cyber security services. However, there are already a number of problems and limitations associated with these solutions since a large number of them require a certain level of technical expertise, paid subscriptions, or even enterprise-level infrastructure.

Phishing scams currently occurring on the internet today are becoming more complex and harder to identify manually by common internet users. Ordinary internet users would likely not recognize fake login pages, suspicious website addresses, and newly created phishing sites. Some of the security solutions for preventing phishing attacks provide a warning sign without indicating the cause of the warning itself.

The purpose of this project is to create a user-friendly and affordable phishing detection system that can analyze URLs, determine phishing scores, and generate phishing alerts and explanations.

### **3. Problem Statement**

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The problem of phishing attacks will not stop rising because most internet users have difficulty recognizing malicious websites. Criminals use misleading URLs, phony domains, and fake web pages to steal personal user information like passwords, credit card numbers, and personal data. Even if there are already many phishing detection systems in existence, not all of these can provide transparent outputs, real-time analysis, and reasonable price for average users. They could also be highly dependent on paid security systems and infrastructures. For these reasons, the need for an intelligent phishing detection system that can analyze URLs, determine phishing scores, and display real-time warnings with explanations is necessary.

### **4. Aim and Objectives**

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#### **4.1 Project Aim**

The aim of this project is to develop a phishing detection system for identify malicious URLs and suspicious websites using python-based technologies and rule base analysis.

#### **4.2 Project Objectives**

- To analyze URLs using phishing detection rules.
- To identify suspicious website characteristics and phishing indicators
- To calculate phishing risk scores based on URL analysis.
- To store scanned URLs and detection result in a database.
- To develop a dynamic web-based user interface for phishing analysis.
- To provide explainable phishing detection report to users.
- To improve cyber security awareness among internet users.

## 5. Methodology

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### 5.1 Development Methodology

The Agile software development methodology will be used for the implementation of the proposed system. Agile methodology supports incremental development, continuous testing, and flexible enhancement of system features. The project will be developed in multiple phases including requirement gathering, system design, development, testing, deployment, and documentation.

### 5.2 Proposed System Overview

This system is an anti-phishing solution written using Python programming language and comes equipped with a browser extension that scans suspicious websites and warns the user about any potential threats. This system employs rule-based phishing analysis methods to detect any suspicious URL patterns, spoofed websites, strange characters, and dangerous website behavior.

The user can scan URLs either using the web interface or the browser extension itself. As the user visits any suspicious website, the browser extension will contact the Python server to conduct the analysis and provide real-time alerts.

The system assigns a phishing risk score to websites along with an explanation of their suspicious behavior.

### 5.3 System Workflow

1. User visits a website or enters a URL.
2. Browser extension captures the URL.
3. Python backend extracts URL features.
4. Detection engine analyzes phishing indicators.
5. Phishing risk score is calculated.
6. Warning notification is displayed.
7. Scan result is stored in the database.

## 5.4 Functional Requirements

Table 5.4: Functional Requirements

| Functional Requirement | Description                                    |
|------------------------|------------------------------------------------|
| URL Input              | Users can enter URLs for scanning              |
| URL Analysis           | System analyzes suspicious URL features        |
| Risk Score Generation  | System calculates phishing risk score          |
| Detection Result       | System displays safe/phishing status           |
| Scan History           | User can view previous scans                   |
| Browser Extension      | Detect suspicious websites directly in browser |
| Admin Dashboard        | Admin can monitor scan statistics              |
| Database Storage       | Scan records are stored in database            |
| Real-Time Warning      | Show warning notifications instantly           |

## 5.5 Non-Functional Requirements

Table 5.5: Non-Functional Requirements

| Non-Functional Requirement | Description                                  |
|----------------------------|----------------------------------------------|
| Security                   | Secure authentication and data handling      |
| Performance                | Fast URL analysis and response               |
| Reliability                | Stable and accurate detection process        |
| Usability                  | User Friendly web interface                  |
| Scalability                | Ability to handle increasing number of scans |
| Availability               | Accessible through web browser               |

## 5.6 User Levels and User Roles

Table 5.6: User levels and user roles

| User Levels   | User Role                                                  |
|---------------|------------------------------------------------------------|
| User          | Perform basic URL scans<br>Scan URLs and view scan history |
| Administrator | Manage reports, monitor statistics, and maintain system    |

## 5.7 Input, Process, and Output

### Inputs

- URLs
- Domain names
- User reports

### Process

- URL feature extraction
- Rule-based phishing analysis
- Risk score calculation
- Database storage

### Outputs

- Safe or phishing detection result
- Phishing risk score
- Warning messages
- Detection explanation
- Scan history records

## 5.8 Use Case

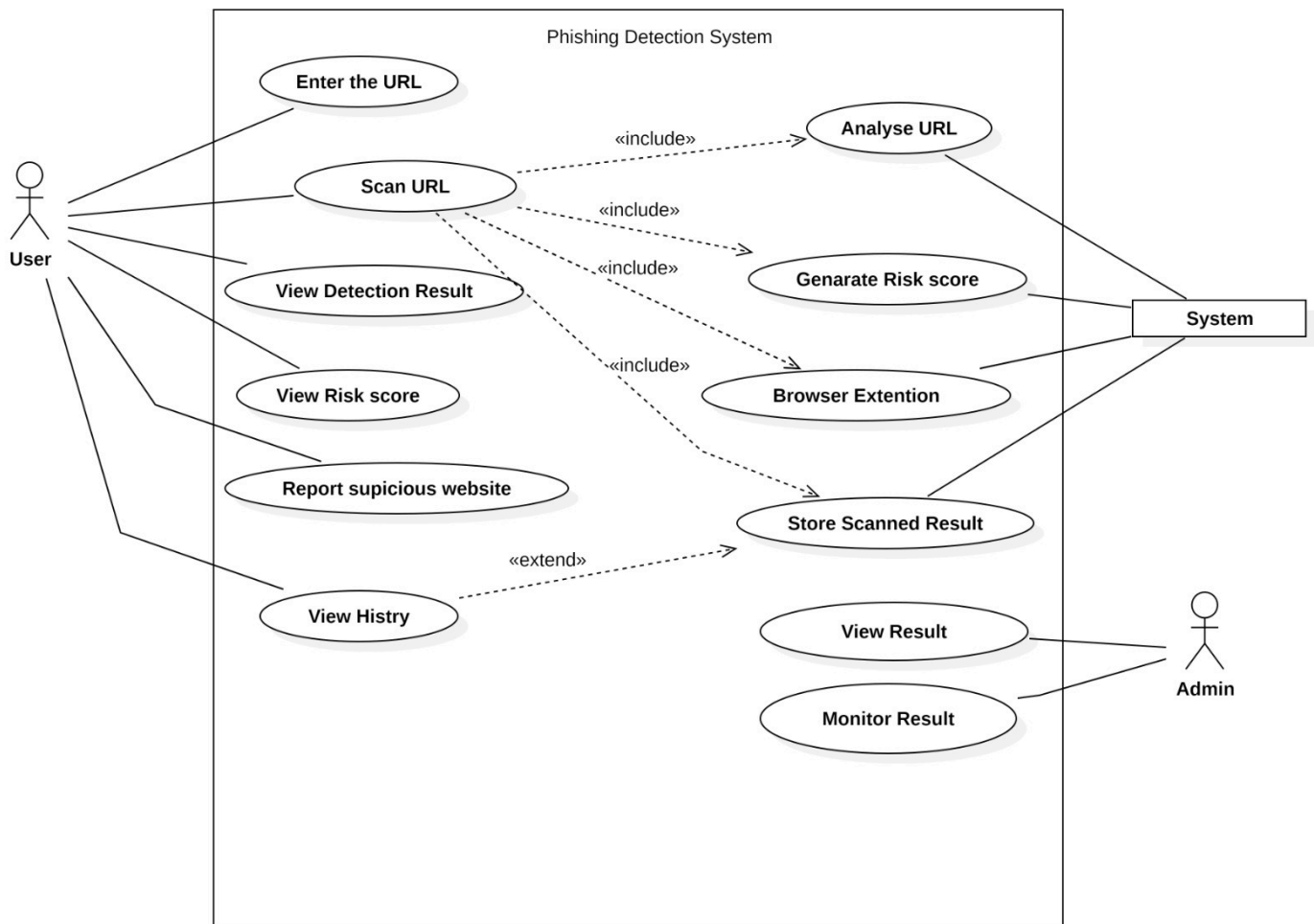


Figure 5.8: Project Time Plan

The Phishing Detection System allows users to enter and scan website URLs to identify potential phishing threats. The system analyses the URL, detects suspicious patterns, and generates a risk score to determine whether the website is safe or harmful. Users can view detection results, check previous scan history, and report suspicious websites for further investigation. The system also supports a browser extension for quick scanning during web browsing and stores scanned results in a database. Additionally, administrators can monitor scanning activities and review detection results to improve system security and phishing prevention.

## 5.9 Technologies Used

Table 5.9: Technologies Used

| Component       | Technology             |
|-----------------|------------------------|
| Frontend        | HTML, CSS, JavaScripts |
| Backend         | Python                 |
| Database        | MongoDB                |
| IDE             | Visual Studio Code     |
| Version Control | GitHub                 |

## 5.10 System Architecture

Replace architecture with:

- User Interface
- Browser Extension
- Python Backend Server
- Phishing Detection Engine
- Database Server

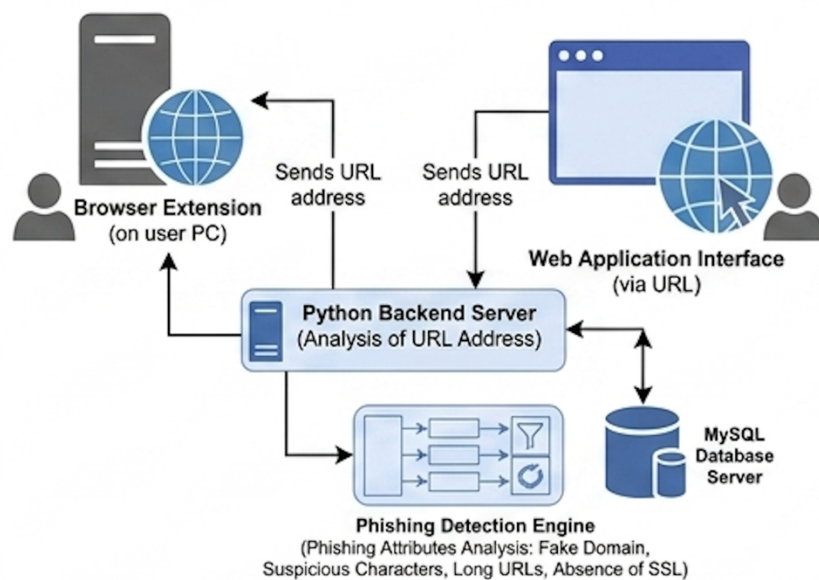


Figure 5.10: System Architecture

System architecture of the proposed system encompasses browser extension, web application interface, Python backend server, detection module for phishing, and MySQL database. In case the user accesses any webpage or types the URL address, then either browser extension or web interface sends the URL address to the Python backend server where analysis of the URL address will be performed. In order to determine whether the URL address is potentially phishing or not, analysis of the phishing attributes of the URL address such as existence of a fake domain name, suspicious characters, very long URLs, or absence of SSL, is performed by the detection module.

## 6. Resource Requirements and Budget

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### 6.1 Hardware Requirements

Table 6.1: Hardware Requirement

| Resource            | Specification      |
|---------------------|--------------------|
| Computer/Laptop     | Intel i5 or higher |
| RAM                 | Minimum 8GB        |
| Storage             | 20GB or higher     |
| Internet Connection | Required           |

## 6.2 Software Requirements

Table 6.2: Software Requirements

| Software                            | Purpose                                           |
|-------------------------------------|---------------------------------------------------|
| Visual Studio Code & GitHub Copilot | Frontend and Backend development                  |
| GitHub                              | Version control                                   |
| Google Chrome                       | System testing                                    |
| MongoDB                             | Database management and Data Storage              |
| Figma                               | UI/UX design, wireframing, and prototype creation |

## 6.3 Budget

Table 6.3: Budget

| Item                   | Estimated Cost    |
|------------------------|-------------------|
| Internet Usage         | LKR 2,000.00      |
| Software Tools         | Free/Open source  |
| Hosting and Deployment | Free tier service |
| Total Estimated Cost   | LKR 2,000         |

## 7. Project Time Plan

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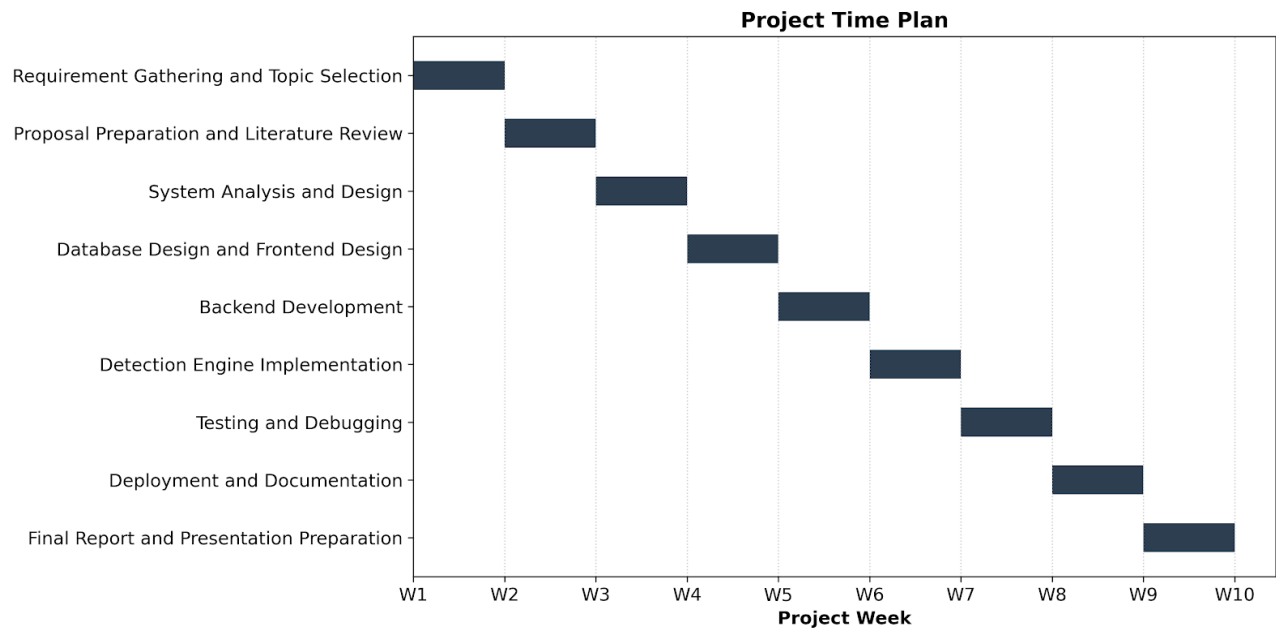


Figure 7.1: Project Time Plan

Table 7.1: Project time plan

| Activity                        | Week    |
|---------------------------------|---------|
| Requirement Gathering           | Week 01 |
| Proposal & Literature Review    | Week 02 |
| Proposal & Literature Review    | Week 03 |
| Database & Frontend Design      | Week 04 |
| Backend Development             | Week 05 |
| Detection Engine Implementation | Week 06 |
| Testing & Debugging             | Week 07 |
| Deployment & Documentation      | Week 08 |
| Final Report & Presentation     | Week 09 |

## 8. Expected Outcome

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The ultimate system will be able to offer a phishing detection framework within the browser that is able to analyze URLs in real-time using the Python tools. Warnings will be notified to users via the browser extension whenever potential threats are identified. The system will generate phishing risk scores and provide warning messages with detailed explanations regarding suspicious URL behavior.

## 9. Future Enhancements

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The proposed system can be further enhanced with the following features:

- Machine learning-based phishing prediction
- QR code phishing detection
- Email phishing analysis
- Mobile application development
- Advanced threat intelligence integration
- Screenshot-based phishing analysis

## 10. References

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- 2] Python Software Foundation , “Python Documentation,” 2026. [Online]. Available: <https://docs.python.org/3/>. [Accessed: May 14, 2026].
- 3] MongoDB, “MongoDB Documentation,” <https://github.com/mongodb/docs-bi-connector/blob/DOCSP-3279/source/index.txt>, 2024. <https://www.mongodb.com/docs/>
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**Group Details:**

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Signature of the Team Members:

| Registration Number | Name                        | Contact Number | Signature |
|---------------------|-----------------------------|----------------|-----------|
| UWU/ICT/23/004      | K. A. N. P. Perera          | 075 681 2635   |           |
| UWU/ICT/23/005      | L. V. C. S. U. Kumari       | 070 159 3628   |           |
| UWU/ICT/23/026      | S. H. Thebuwana             | 075 998 3411   |           |
| UWU/ICT/23/080      | A. W. D. H. T. Jayawickrama | 075 755 0779   |           |
| UWU/ICT/23/081      | S. A. N. Rathnasinghe       | 076 031 4524   |           |

Date: .....

Recommendation of Supervisor

Name: Mr. V. Dishankan

Department/Organization: Department of Information and Communication Technology

Signature: .....